

Assignment 3 - Case Study Part 3

Abe Tayo Noah. Nguyen Hai Yen.

David Enrique Garcia Olvera. Nizami, Muhammad Abu Naim

Master of Data Analytics, University of Niagara Falls Canada

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1. Statement of Purpose

The use of credit cards has grown rapidly in recent years, enabling customers to conduct a wide variety of financial transactions with convenience and flexibility. However, it also increases the risk of credit card defaults for banks when customers are unable to repay the amount they have borrowed from the bank. This leads to banks facing financial losses, higher operating costs, and more complex debt collection activities.

To address this challenge, financial institutions increasingly rely on data-driven methods to assess customer risk and anticipate potential defaults. This project uses a real-world credit card dataset from a Taiwanese bank to examine customer spending patterns, repayment behaviors, and demographic characteristics. By identifying the factors strongly associated with default, the project aims to build a predictive model that can estimate the likelihood of a customer defaulting in the following month. Such predictive insights support better decision-making in credit risk management, including credit limit planning, targeted monitoring, and early intervention strategies.

The project was carried out through several steps in data analysis, such as formulating research questions about which factors or behaviors affect customers' credit card defaults. From there, hypotheses were developed, data exploration was performed, and statistical models were built to clarify these relationships. Tools used in the project included Python, SQL for data exploration, Python and Tableau for data visualization, and Python for model building.

2. Scope of the Project

This project focuses exclusively on descriptive and predictive analytics using the Taiwanese credit card default dataset. The scope includes data understanding, visualization,

hypothesis testing, and the development and evaluation of predictive models for default classification. The analysis centers on identifying key behavioral and demographic factors that influence credit card default risk and comparing the predictive power of different feature groups, particularly demographics versus payment behavior.

The project does not include prescriptive analytics, cost optimization, intervention strategies, or real-time credit scoring system design. Instead, it provides a structured analytical foundation that supports future extensions into risk mitigation and operational decision-making. The findings generated are limited to the dataset provided and do not generalize to other financial institutions without further validation.

3. Background Research and Literature

Many studies have shown that data analytics plays a crucial role in predicting credit card default risk and improving the risk management effectiveness of financial institutions. Costa E Silva, Eliana; Lopes, Cristina; Correia, Aldina; and Faria, S. (2020) developed a logistic regression model to predict the default risk of borrowers in Portugal. The model identified key factors affecting the likelihood of default, achieved nearly 90% accuracy, and passed multiple statistical validation tests. The study highlights the significant role of customer behavioral data in enhancing credit analytics models.

Similarly, Zhu, L., Qiu, D., Ergu, D., Ying, C., and Liu, K. (2019) developed a predictive model for loan defaults on the Lending Club platform using the Random Forest algorithm. The results demonstrated that the decision tree-based model provided higher classification accuracy than traditional methods and effectively handled nonlinear relationships and complex interactions among variables. This research emphasizes that advanced machine learning techniques can substantially improve the predictive capability of financial institutions.

Together, these studies indicate that applying data analytics techniques from classical statistical models like logistic regression to modern machine learning methods such as Random Forest is essential for enhancing the accuracy of risk prediction and developing early warning systems in the credit domain.

4. Problem Definition

To manage credit risk effectively, banks need to detect early those customers who are at high risk of defaulting on their credit card debt, using data available when customers apply for or use their cards. If banks misjudge the risk, it can lead to increased default rates and high debt collection costs when accepting high-risk customers. Conversely, if they reject low-risk customers, the bank will lose revenue and market share.

Therefore, the business problem is how banks can use historical data to predict which customers are likely to default in the next period. The project focuses on a binary classification problem, which means predicting the outcome with two values: 1 for default and 0 for no default. This approach aligns with current credit risk models and helps evaluate the effectiveness of the predictive model in a clear and practical way.

5. Research Questions

To guide our analysis, we propose the following research questions:

RQ1: Which factors (such as age, education, credit limit, and payment behavior) most affect a customer's likelihood of defaulting on their credit card?

RQ2: Are recent payment behaviors (such as late payments) more predictive of default than static demographic characteristics?

6. Hypotheses Formulation

From the above research questions, the project develops the following research hypotheses:

- Hypothesis 1 – Credit Limit
 - H0: $\mu_{\text{default}} = \mu_{\text{non_default}}$
 - H1: $\mu_{\text{default}} < \mu_{\text{non_default}}$
- Hypothesis 2 – Payment Delays
 - H0: Probability of default is independent of PAY_0–PAY_6
 - H1: Clients with higher payment delays have higher probability of default
- Hypothesis 3 – Bill-to-Payment Ratios:
 - H0: Bill-to-payment ratios have no effect on default probability
 - H1: Higher ratios are associated with higher default risk
- Hypothesis 4 – Demographics
 - H0: Default rates are the same across demographic groups
 - H1: Default rates differ significantly across demographic groups

7. Methodology

We use t-tests, chi-square tests, and logistic regression in Python to test our hypotheses. Specifically, t-tests are applied to continuous variables (LIMIT_BAL, AGE, BILL_AMT1–6, PAY_AMT1–6) to compare means between defaulting and non-defaulting clients. For categorical variables such as SEX, EDUCATION, and MARRIAGE, chi-square tests are conducted to examine differences in default rates across groups. Additionally, logistic regression and random forest are employed to assess the impact of each variable on the probability of default and to evaluate the direction and statistical significance of the

coefficients. These methods are chosen because they are suitable for the data types and align with the research objective of hypothesis testing. The implementation is done in Python using libraries such as pandas, scipy.stats, and statsmodels.

8. Data Collection

The dataset used in this project, “*Default of Credit Card Clients*,” is publicly available from the UCI Machine Learning Repository (Yeh, 2009). It contains information on 30,000 credit card clients in Taiwan, capturing both demographic characteristics and financial behavior over a six-month period.

Key Features:

- Number of Observations: 30,000 clients
- Number of Variables: 23 predictors plus 1 binary target variable
- Target Variable: default.payment.next.month (1 = default, 0 = non-default)
- Data Type: Mostly numeric (integer/continuous); no missing values reported

This dataset offers a comprehensive foundation for analyzing credit risk patterns and developing predictive models.

Data Acquisition and Preprocessing with Python

The dataset can be downloaded as an Excel file from the UCI repository or Data Science Dojo. We used Python for the initial stages of Preprocessing.

```
import pandas as pd

df = pd.read_excel('default_of_credit_card_clients.xls', header=1)

display(df.head())

# Check for null values
```

```
null_counts = df.isnull().sum()
print("Null values per column:")
display(null_counts[null_counts > 0])

# Check for duplicate rows
duplicate_rows = df.duplicated().sum()
print(f"\nNumber of duplicate rows: {duplicate_rows}")
```

Python was used to load the data into the working environment using `pandas.read_excel()`. The dataset was carefully examined for data quality issues. No missing values or duplicate rows were found, indicating that the dataset is complete and consistent. Additionally, the target variable was renamed for clarity, ensuring readability and consistency in subsequent analyses.

9. Data Understanding

Before building predictive models, it is essential to understand the structure and characteristics of the dataset. This phase focuses on variable definitions, descriptive statistics, and the distribution of the target variable.

9.1. Variable Overview

According to the UCI documentation, the dataset includes the following categories of variables:

Demographic Variables

- SEX: Gender (1 = male, 2 = female)
- EDUCATION: Education level (1 = graduate school, 2 = university, 3 = high school, 4 = others)
- MARRIAGE: Marital status (1 = married, 2 = single, 3 = others)
- AGE: Age in years

Credit and Account Variables

- LIMIT_BAL: Total credit limit granted (in NT dollars), including personal and supplementary credit

Payment History Variables

- PAY_0 to PAY_6: Repayment status from April to September 2005. Negative values indicate on-time payments; positive values indicate months of delay.

Bill Statement Variables

- BILL_AMT1 to BILL_AMT6: Monthly bill amounts over six months

Payment Amount Variables

- PAY_AMT1 to PAY_AMT6: Actual payment amounts over six months

Target Variable

- default: Binary indicator of default in the next month (1 = default, 0 = non-default)

9.2.Initial Descriptive Statistics

Using Python, we compute summary statistics mean, median, standard deviation, minimum, and maximum for key numeric variables such as LIMIT_BAL, AGE, BILL_AMT1–6 and PAY_AMT1–6.

Table 9.2.1

Summary statistics

	LIMIT_BAL	AGE	BILL_AMT1	BILL_AMT2	BILL_AMT3	BILL_AMT4	BILL_AMT5	BILL_AMT6	PAY_AMT1	PAY_AMT2	PAY_AMT3	PAY_AMT4	PAY_AMT5	PAY_AMT6
count	30000.000000	30000.000000	30000.000000	30000.000000	3.000000e+04	30000.000000	30000.000000	30000.000000	30000.000000	3.000000e+04	30000.000000	30000.000000	30000.000000	30000.000000
mean	167484.322667	35.485500	51223.330900	49179.075167	4.701315e+04	43262.948967	40311.400967	38871.760400	5663.580500	5.921163e+03	5225.68150	4826.076867	4799.387633	5215.502567
std	129747.661567	9.217904	73635.860576	71173.768783	6.934939e+04	64332.856134	60797.155770	59554.107537	16563.280354	2.304087e+04	17606.96147	15666.159744	15278.305679	17777.465775
min	10000.000000	21.000000	-165580.000000	-69777.000000	-1.572640e+05	-170000.000000	-81334.000000	-339603.000000	0.000000	0.000000e+00	0.000000	0.000000	0.000000	0.000000
25%	50000.000000	28.000000	3558.750000	2984.750000	2.666250e+03	2326.750000	1763.000000	1296.000000	1000.000000	8.330000e+02	390.000000	296.000000	252.500000	117.750000
50%	140000.000000	34.000000	22381.500000	21200.000000	2.008850e+04	19052.000000	18104.500000	17071.000000	2100.000000	2.009000e+03	1800.000000	1500.000000	1500.000000	1500.000000
75%	240000.000000	41.000000	67091.000000	64006.250000	6.016475e+04	54506.000000	50190.500000	49198.250000	5006.000000	5.000000e+03	4505.000000	4013.250000	4031.500000	4000.000000
max	1000000.000000	79.000000	964511.000000	983931.000000	1.664089e+06	891586.000000	927171.000000	961664.000000	873552.000000	1.684259e+06	896040.000000	621000.000000	426529.000000	528666.000000
median	140000.000000	34.000000	22381.500000	21200.000000	2.008850e+04	19052.000000	18104.500000	17071.000000	2100.000000	2.009000e+03	1800.000000	1500.000000	1500.000000	1500.000000

The given table shows some key observations:

- The average age of clients is approximately 35–36 years, with most falling between their early 20s and early 40s.
- The dataset is imbalanced: roughly 22% of clients defaulted ($\approx 6,636$ out of 30,000).
- The LIMIT_BAL variable is right-skewed, indicating that while many clients have modest credit limits, a small subset holds significantly higher limits.

These insights help shape our modeling strategy, especially in handling class imbalance and variable transformations.

9.3.Target Variable Distribution

We examine the distribution of the target variable default to confirm the class imbalance:

Table 9.3.1

Distribution of 'default' target variable

Default	Count	Percentage
0	23364	77.88
1	6636	22.12

About 22% of clients are defaulters and 78% are non-defaulters, creating a significant class imbalance. This imbalance is important to consider during model evaluation, since

accuracy alone may be misleading especially when assessing the model's ability to correctly identify defaulters.

9.4. SQL Queries

Although the primary analysis for this project is conducted in Python, we also use SQL during the initial exploratory stages to cluster data and observe general patterns.

- Analyze credit card default rate based on default_payment_next_month column

- o SELECT

- CASE WHEN default_payment_next_month = 1 THEN 'Default'
 - ELSE 'No Default' END AS status,
 - COUNT(*) AS count,
 - ROUND(COUNT(*) * 100.0 / (SELECT COUNT(*) FROM credit_card), 2) AS percent
 - FROM credit_card
 - GROUP BY status;

Table 9.4.1

Default Card Rate

Status	Count	Percent
Default	6636	22.12
No Default	23364	77.88

- Analyze default rate by education and marital status

- o SELECT EDUCATION, MARRIAGE, AVG(AGE) AS avg_age,
 - AVG(default_payment_next_month) AS default_rate
 - FROM credit_card

GROUP BY EDUCATION, MARRIAGE

order by default_rate desc;

Table 9.4.2

Top 5 Default Rate by Education and Marital Status

Education	Marriage	Avg. Age	Default Rate
2	3	40.8025	0.2716
3	1	43.0825	0.2635
3	3	46.6408	0.2621
1	3	41.98	0.26
2	1	38.4991	0.2435

10. Statistical Test

We conducted t-tests for numeric variables and chi-square tests for categorical variables to evaluate the hypotheses regarding default risk.

10.1. Results of Statistical Tests**Table 10.1.1**

T-Test Results (Numeric Variables)

Variable	t-stat	p-value	Significance
LIMIT_BAL	-28.951588	3.36E-178	Significant
PAY_AMT1	-17.512542	3.55E-68	Significant
PAY_AMT2	-14.817166	1.88E-49	Significant
PAY_AMT4	-12.222113	3.37E-34	Significant
PAY_AMT3	-11.893461	1.78E-32	Significant

PAY_AMT5	-11.245665	3.24E-29	Significant
PAY_AMT6	-11.058006	2.59E-28	Significant
BILL_AMT1	-3.397756	6.82E-04	Significant
BILL_AMT3	-2.459066	1.39E-02	Significant
BILL_AMT2	-2.446597	1.44E-02	Significant
AGE	2.319521	2.04E-02	Significant
BILL_AMT4	-1.758788	7.86E-02	Not Significant
BILL_AMT5	-1.162327	2.45E-01	Not Significant
BILL_AMT6	-0.93021	3.52E-01	Not Significant

The t-test results comparing continuous variables between default and non-default groups indicate that LIMIT_BAL and PAY_AMT1–6 have a strong impact on default risk, with p-values far below 0.001. Clients with higher credit limits are significantly less likely to default, while payment amounts consistently influence the probability of default. Some bill amounts, specifically BILL_AMT1–3, show statistically significant differences between groups, although the effect is weaker, while BILL_AMT4–6 are not statistically significant. Age has a small but statistically significant influence. Overall, these continuous variables provide meaningful insights into factors affecting default.

Table 10.1.2

Chi-Square Test Results (Categorical Variables)

Variable	chi2	p-value	Significance
EDUCATION	163.22	1.23E-32	Significant
SEX	47.71	4.94E-12	Significant

MARRIAGE	35.66	8.83E-08	Significant
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The chi-square tests for categorical variables show that EDUCATION, SEX, and MARRIAGE are all significantly associated with default status. Education level has a very strong effect on default risk, while gender and marital status also exhibit significant, though comparatively smaller, influence. These results indicate that demographic factors play an important role in predicting default.

Table 10.1.3

Chi-Square Test Results (Payment Status PAY_0–PAY_6)

Variable	chi2	p-value	Significance
PAY_0	5365.96	0	Significant
PAY_2	3474.47	0	Significant
PAY_3	2622.46	0	Significant
PAY_4	2341.47	0	Significant
PAY_5	2197.69	0	Significant
PAY_6	1886.84	0	Significant

All payment indicators have a very strong effect on default risk. Clients with higher late payments are more likely to default.

10.2. Hypothesis Testing

The results of our statistical analyses support all four hypotheses:

- Credit Limit (H1): Clients with higher credit limits (LIMIT_BAL) are significantly less likely to default, confirmed by a very low p-value in the t-test and a negative logistic regression coefficient.

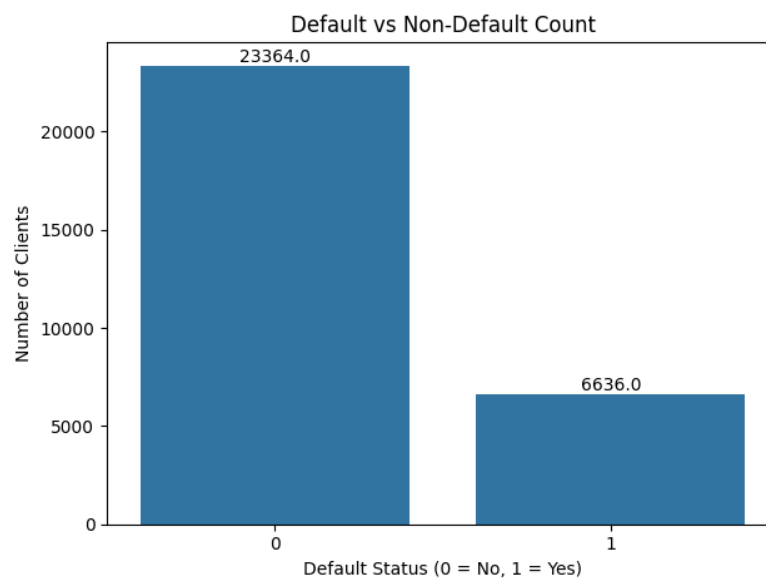
- Payment Delays (H2): Clients with higher recent payment delays (PAY_0–PAY_6) have a significantly higher probability of default, supported by chi-square tests and regression coefficients.
- Bill-to-Payment Ratios (H3 Higher ratios of bill amounts to payments (BILL_AMT1–3 / PAY_AMT1–6) are associated with increased default risk. BILL_AMT4–6 are not statistically significant.
- Demographics (H4): Default rates differ significantly across demographic groups (SEX, EDUCATION, MARRIAGE), as shown by chi-square tests.

11. Data Visualization

We explored the data and visualized key information to draw out key insights, using Python and Tableau as analytical tools.

Figure 11.1

Default vs Non-Default Count

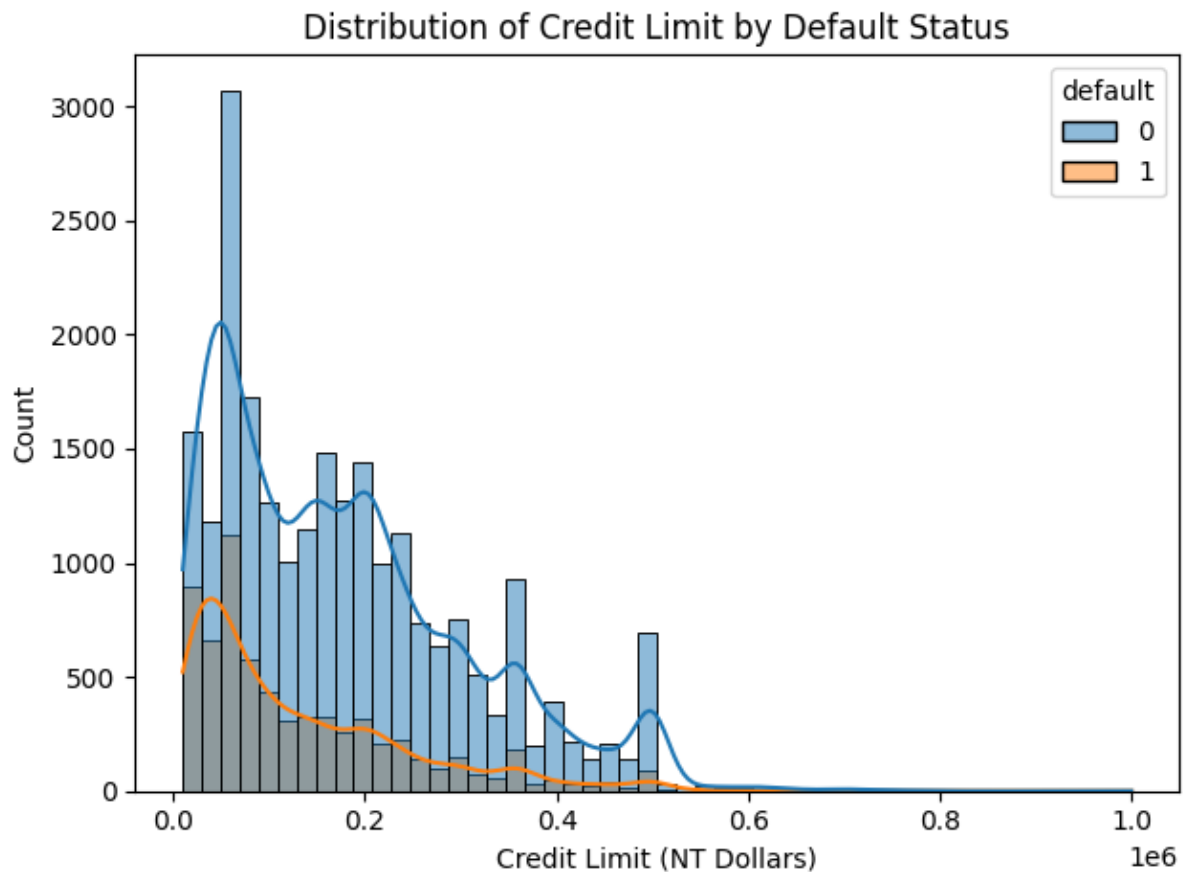


The bar chart indicates that 23,364 clients did not default, and 6,636 clients did default on their credit card payments. This confirms a clear imbalance, with non-defaulters heavily

outnumbering defaulters. Such an imbalance must be considered during analysis and model development, as models trained on skewed data may become biased toward the majority class.

Figure 11.2

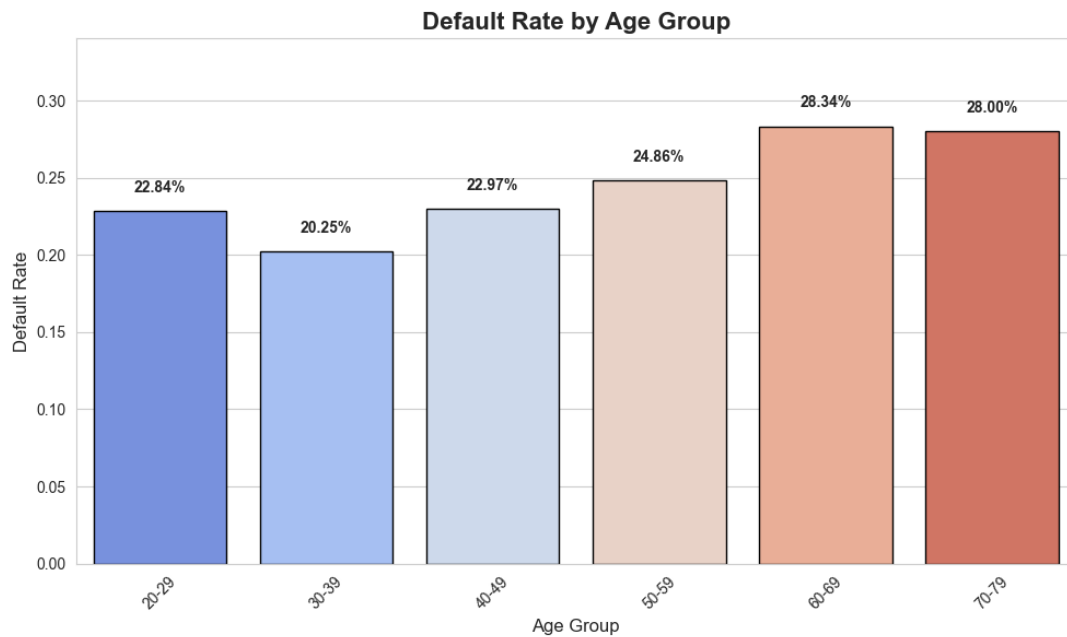
Credit Limit Distribution by Default Status



The distributions of credit limits for defaulters and non-defaulters overlap significantly, meaning clients with similar credit limits can either default or not. Non-defaulters generally make up a larger portion, especially at low to mid credit limits. Defaulters appear across all credit limits but are more concentrated at lower levels; as credit limits increase, non-defaulters become more dominant. Overall, credit limit is a relevant factor but not a definitive indicator of whether a client will default.

Figure 11.3

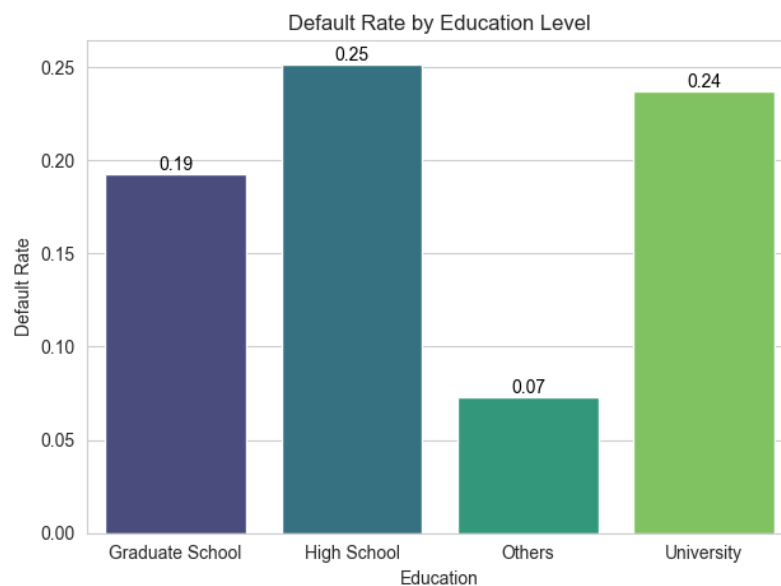
Default Rate by Age Group



The bar chart shows that clients aged 30–39 have the lowest default rate (around 20%). The 20–29 and 40–49 groups have similar rates, about 22–23%. Default rates increase noticeably from the 50+ age groups, reaching around 25–28%. Overall, clients in their thirties are the least likely to default, while older age groups show a higher likelihood of defaulting.

Figure 11.4

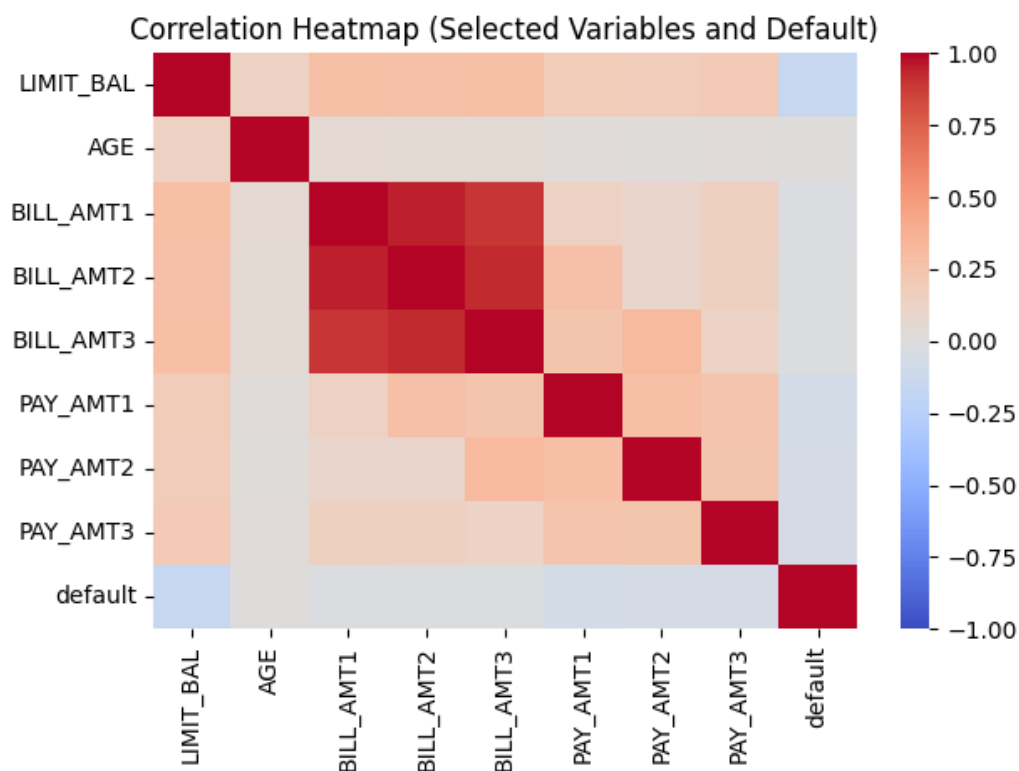
Default Rate by Education Level



The chart illustrates the default rates by education level. Among the standard groups, those with a graduate degree have the lowest default rate at 19.23%, followed by university graduates at 23.73% and high school graduates at 25.16%. Overall, higher education tends to be associated with lower default risk, although some outlier groups may require further investigation.

Figure 11.5

Correlation Heatmap of Selected Variables

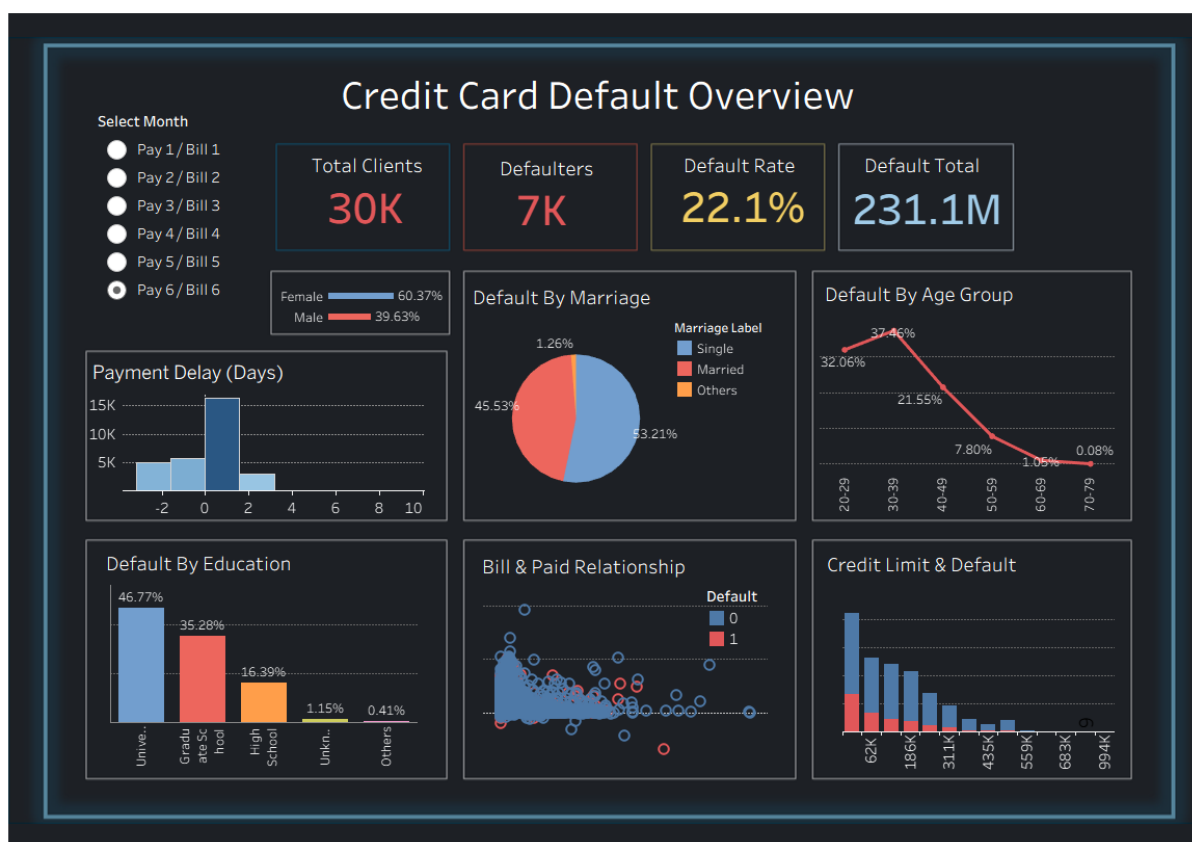


The heatmap plot shows that credit limit (LIMIT_BAL) and payment amounts (PAY_AMT) have a slight relationship with default, meaning clients with higher limits and larger payments are less likely to default. Age (AGE) and the first few months' bill amounts (BILL_AMT1–3) have very weak effects, providing little predictive power for default. The BILL_AMT columns are highly correlated with each other, indicating that monthly bill

amounts tend to move together, while LIMIT_BAL has a moderate correlation with the bills. The PAY_AMT columns also show weak positive correlations among themselves. Overall, some variables have weak links to default, but the strong correlations among bill amounts should be considered when building predictive models.

Figure 11.6

Dashboard Overview



Additionally, the Tableau dashboard's payment timing analysis adds an important insight: the majority of customers pay almost on time, with delays centered around -2 to 2 days, meaning they can pay early or on time. However, there is a small group of customers who pay significantly late, up to 8–10 days or more, and this group is often at higher than average risk of default, especially when combined with risk factors such as young age or low credit limit.

These customers should be monitored closely, as payment timing is an important metric for assessing and managing default risk.

12. Model Building

To evaluate the influence of different factors on the probability of default and to answer the research questions, we developed several predictive models using Python. Logistic Regression was selected as the primary modeling technique because it is well-suited for binary classification problems and provides interpretable coefficients that indicate the direction and strength of each predictor's relationship with default risk. Three logistic regression models were constructed to isolate the predictive contribution of different feature groups: a full model incorporating all demographic, repayment, billing, and payment amount variables; a demographic-only model; and a behavior-focused model emphasizing repayment status and recent financial activity. This design allows comparison of how strongly payment behavior predicts default relative to demographic characteristics and ensures alignment with both research questions. Although other algorithms such as Random Forest or Gradient Boosted Trees could capture nonlinear patterns, Logistic Regression was prioritized for transparency, regulatory alignment, and interpretability in the context of credit risk assessment.

12.1 Data Preparation

In the first step, we performed feature selection based on our research objectives, considering three sets of variables: all features, demographics only, and behavior plus credit variables. To reduce skewness in certain variables, particularly LIMIT_BAL and PAY_AMT1–6, we applied a log transformation ($\log_{1p}$). We also addressed missing and infinite values by replacing inf with NaN and removing rows containing NaN in any of the necessary variables. The target variable for our modeling was set as DEFAULT.

12.2. Model Building

12.2.1 Logistic Regression

We split the dataset into training and testing sets using a 70–30 ratio, stratifying by the target variable to maintain a similar proportion of default and non-default cases in both sets. Feature scaling was applied using `StandardScaler` to ensure all features are on the same scale, which helps the Logistic Regression model converge more efficiently. The Logistic Regression model was configured with a maximum of 1000 iterations to ensure convergence, and `class_weight='balanced'` was used to handle class imbalance, as default cases only accounted for approximately 20% of the data. We used the `lbfgs` solver, which is suitable for small to medium-sized datasets.

```
def run_logistic_regression(stat_df, feature_list, y, model_name):
    X = stat_df[feature_list]

    # Train-test split
    X_train, X_test, y_train, y_test = train_test_split(
        X, y, test_size=0.3, stratify=y, random_state=42
    )

    # Scale features
    scaler = StandardScaler()
    X_train_scaled = scaler.fit_transform(X_train)
    X_test_scaled = scaler.transform(X_test)

    # Logistic Regression with imbalance handling
    log_reg = LogisticRegression(
        max_iter=1000,
        class_weight='balanced',
        solver='lbfgs'
    )

    # Fit model
    log_reg.fit(X_train_scaled, y_train)

    # Predictions
    y_pred = log_reg.predict(X_test_scaled)
    y_prob = log_reg.predict_proba(X_test_scaled)[:, 1]
```

```

# Evaluation metrics
metrics = {
    'Accuracy': accuracy_score(y_test, y_pred),
    'Precision': precision_score(y_test, y_pred),
    'Recall': recall_score(y_test, y_pred),
    'F1 Score': f1_score(y_test, y_pred),
    'ROC-AUC': roc_auc_score(y_test, y_prob)
}

print(f"\n===== {model_name} =====")
print("Accuracy :", metrics['Accuracy'])
print("Precision:", metrics['Precision'])
print("Recall    :", metrics['Recall'])
print("F1 Score  :", metrics['F1 Score'])
print("ROC-AUC   :", metrics['ROC-AUC'])

print("\nConfusion Matrix:\n", confusion_matrix(y_test, y_pred))
print("\nClassification Report:\n", classification_report(y_test,
y_pred))

# Coefficients
coef_df = pd.DataFrame({
    'Feature': feature_list,
    'Coefficient': log_reg.coef_[0]
})
coef_df['Abs_Coefficient'] = coef_df['Coefficient'].abs()
coef_df = coef_df.sort_values(by='Abs_Coefficient',
ascending=False)

print("\nTop coefficients (by absolute value):")
print(coef_df[['Feature', 'Coefficient']].head(15))

return log_reg, scaler, coef_df, metrics, y_test, y_pred, y_prob

```

12.2.2 Random Forest Classifier

We continued by building a Random Forest model to compare it with Logistic Regression. The data was split into training and testing sets using a 70–30 ratio and stratified in the same way as before. Unlike Logistic Regression, no feature scaling was applied in this model because Random Forest is tree-based.

The model was trained with 300 decision trees and the parameter `class_weight='balanced'` to address data imbalance, as default cases accounted for only about 20% of the dataset. After training, we evaluated the model using Accuracy, Precision, Recall, F1-score, and ROC-AUC, and also computed the confusion matrix and classification report. In addition, feature importance values were extracted from the Random Forest model to identify the variables that contributed most to the predictions.

```
def run_random_forest(stat_df, feature_list, y, model_name):
    X = stat_df[feature_list]

    # Train-test split
    X_train, X_test, y_train, y_test = train_test_split(
        X, y, test_size=0.3, stratify=y, random_state=42
    )

    # Random Forest
    rf = RandomForestClassifier(
        n_estimators=300,
        max_depth=None,
        class_weight='balanced',
        random_state=42
    )
    rf.fit(X_train, y_train)

    # Predictions
    y_pred = rf.predict(X_test)
    y_prob = rf.predict_proba(X_test)[:, 1]

    # Metrics
    metrics = {
        'Accuracy': accuracy_score(y_test, y_pred),
        'Precision': precision_score(y_test, y_pred),
        'Recall': recall_score(y_test, y_pred),
        'F1 Score': f1_score(y_test, y_pred),
        'ROC-AUC': roc_auc_score(y_test, y_prob)
    }

    # Feature importance
    feat_imp = pd.DataFrame({
        'Feature': feature_list,
        'Importance': rf.feature_importances_
    }).sort_values(by='Importance', ascending=False)
```

```

print(f"\n===== {model_name} =====")
print("Metrics:", metrics)
print("\nConfusion Matrix:\n", confusion_matrix(y_test, y_pred))
print("\nTop Features:\n", feat_imp.head(15))

# Return all outputs
return rf, feat_imp, metrics, y_test, y_pred, y_prob

```

13. Model Evaluation

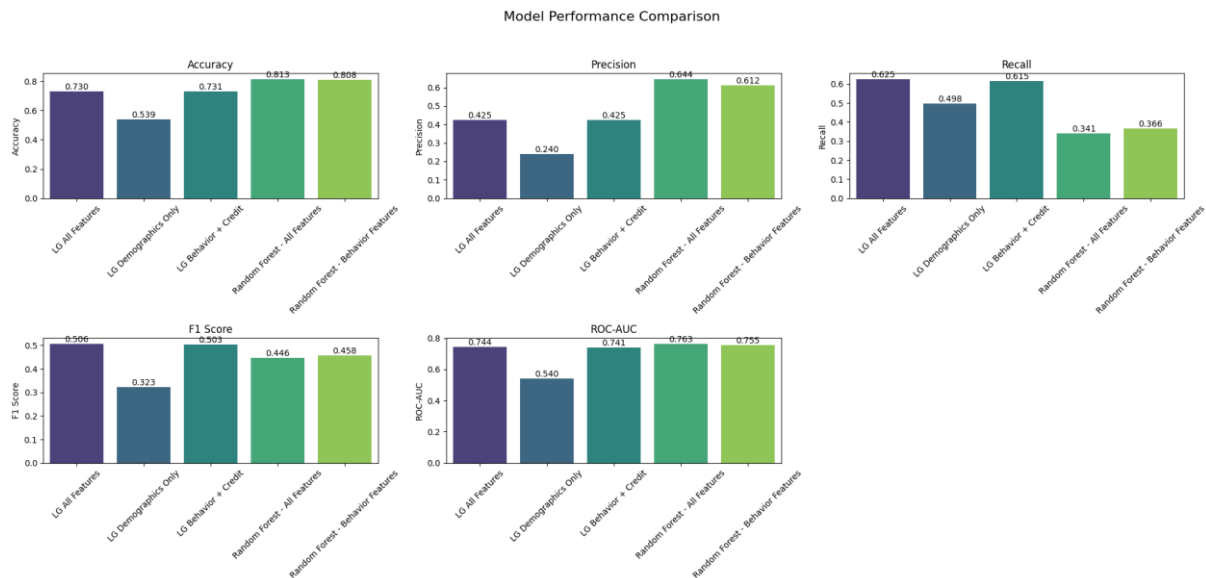
13.1. Model Selection

We evaluated both the Logistic Regression and Random Forest models using multiple metrics to assess their predictive performance on default risk. The evaluation included overall Accuracy, Precision, Recall, F1-score, and ROC-AUC, as well as confusion matrices to provide a detailed view of prediction errors. For Logistic Regression, feature importance was assessed based on model coefficients, while for Random Forest, feature importance was determined from the Gini importance values extracted from the trained trees.

Table 13.1

Model Evaluation Metrics

Model	Variables	Accuracy	Precision	Recall	F1 Score	ROC-AUC
Logistic Regression	All Features	0.73	0.43	0.62	0.51	0.74
	Demographics Only	0.54	0.24	0.5	0.32	0.54
	Behavior + Credit	0.73	0.43	0.61	0.5	0.74
Random Forest	All Features	0.81	0.64	0.34	0.45	0.76
	Behavior Features	0.81	0.61	0.37	0.46	0.76

Figure 13.1*Model Evaluation Metrics Comparison*

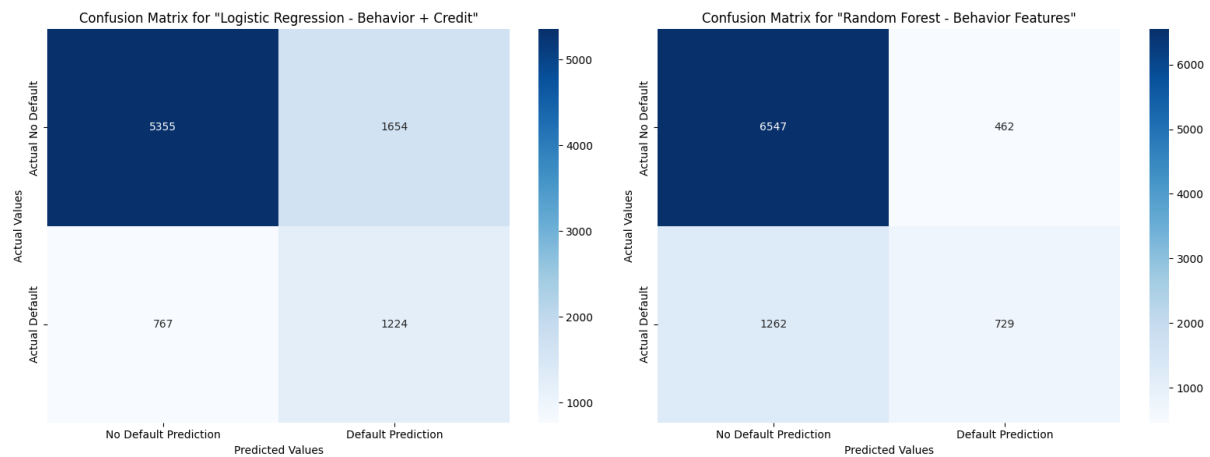
The model using only demographic information with Logistic Regression produces the poorest results. The model achieves an accuracy of only 0.54 and an ROC-AUC of 0.54, indicating that using only demographic features (GENDER, EDUCATION, MARRIAGE, AGE) is hardly effective in assessing default risk. Therefore, using demographic variables alone is insufficient to predict the likelihood of default.

In contrast, combining payment behavior and credit-related features significantly improves performance in both models. In particular, the Random Forest model using only behavioral features achieves comparable performance, with an Accuracy of 0.81, Precision 0.64, Recall 0.34, F1 Score 0.45, and ROC-AUC 0.76. This shows that payment behavior and credit features are the most important predictors.

Thus, based on its performance and simplicity, the Random Forest model using behavioral features appears to be a strong candidate. However, we will further examine its performance through the confusion matrix before making a final conclusion.

Figure 11.1.1

Confusion Matrix

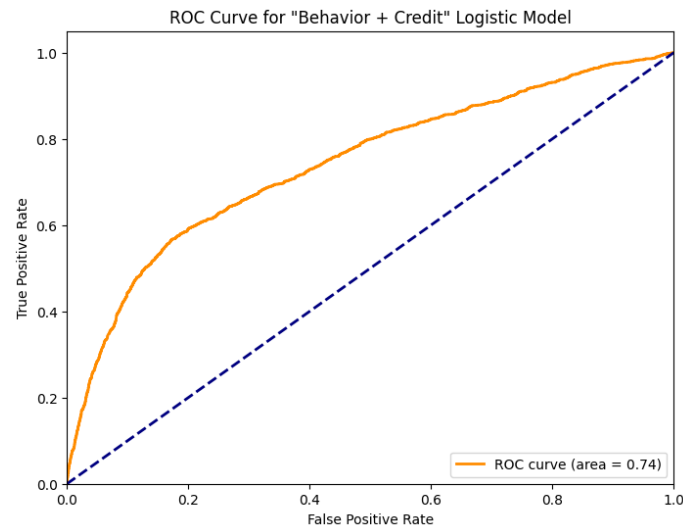


Comparing the performance of the two default prediction models shows that Logistic Regression is more sensitive in detecting defaulting customers (Recall 61%) but generates more false alarms (FP 1,654). In contrast, Random Forest identifies non-defaulting customers more accurately (TN 6,547) and reduces false alarms, but misses more defaulting customers (Recall 36.6%). Given the results of performance metrics and our priority of maximizing risk detection, we select Logistic Regression as the final model for deployment.

13.2. Behavior + Credit Logistic Regression model

Figure 13.1.2

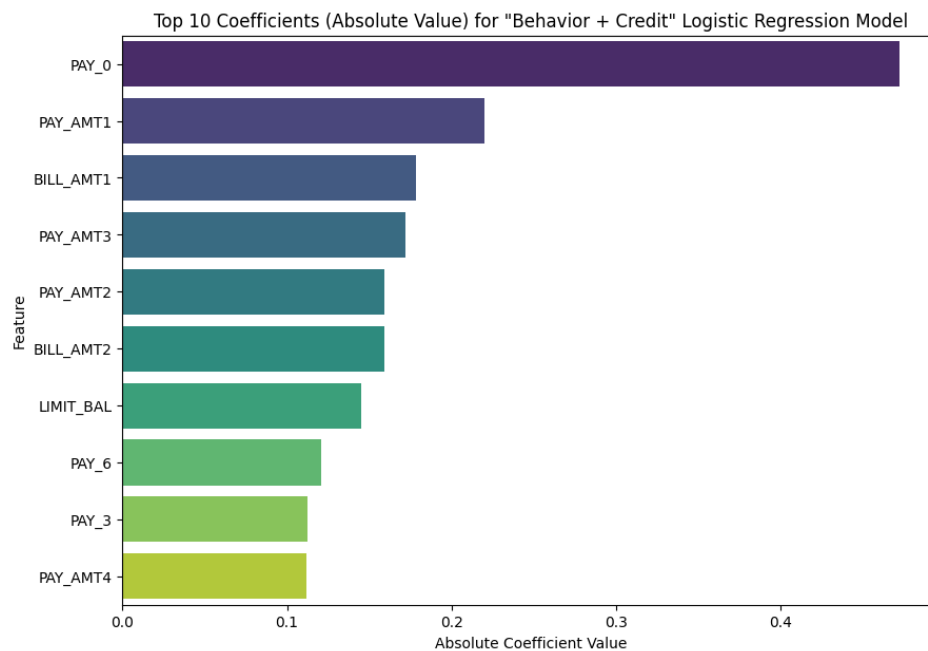
ROC & AUC



The model's AUC is 0.74, indicating a 74% chance of correctly ranking a defaulting customer higher than a non-defaulting one. This shows good discriminative ability.

Figure 13.1.3

Feature Importance



The feature importance analysis for the "Behavior + Credit" model highlights that past payment behavior and credit exposure are the most critical predictors of default. The most

influential feature is PAY_0, representing the most recent payment status, where higher values indicating delayed payments strongly increase default risk. Other important features include PAY_AMT1 (amount paid in month 1), BILL_AMT1 (bill amount in month 1), PAY_AMT2, PAY_AMT3, PAY_6, PAY_3, PAY_AMT4, and LIMIT_BAL, which collectively show that both payment history and credit limits significantly affect the likelihood of default. Conversely, demographic features contribute little to the model, reinforcing that behavioral and credit-related variables drive predictive power.

14. Insights & Recommendations

The Logistic Regression model using only demographic features (GENDER, EDUCATION, MARRIAGE, AGE) gave a very poor result, with Accuracy 0.54 and ROC-AUC 0.54. This result supports H4, showing that single demographic variables do not have strong discriminant value in risk prediction.

Although Random Forest achieved higher overall accuracy and reduced false positives (TN 6.547), this model missed many customers who actually defaulted (Recall 36.6%). In contrast, Logistic Regression is more sensitive in detecting defaulters (Recall 61%) but generates more false alarms (FP 1.654). Since the business objective is to minimize the number of missed risky customers, we choose Logistic Regression as the final model to deploy.

Based on these results, future models should prioritize behavioral and credit features, which can be used independently or optimized for weighting. Enhanced feature engineering related to modeling payment behavior over time, credit usage, and spending trends can further improve accuracy and practical application value.

15. Business Impact & Conclusion

The results of this project have direct implications for financial institutions in managing credit risk. Banks can identify high-risk customers early to reduce potential losses by prioritizing the use of behavioral and credit-related data. The model also enables optimization of credit limits, improves operational efficiency by focusing monitoring efforts on high-risk clients, and supports strategic decision-making based on data.

Behavioral and credit-related factors, such as past payment history, credit utilization, and bill-to-payment ratios, were identified as the strongest predictors of default, while demographic characteristics contributed very little to predictive accuracy. Although Logistic Regression generates more false positives than Random Forest, it is more effective at detecting high-risk customers, aligning with the business objective of minimizing missed risky cases. Future improvements could focus on advanced feature development, including temporal analysis of payment behavior and spending trends, to further enhance predictive capability.

In conclusion, this project provides a practical, data-driven framework for credit risk analysis, supporting both tactical and strategic decision-making within financial institutions, helping banks minimize potential losses and optimize customer management.

16. References

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